NATIONAL RESEARCH UNIVERSITY HIGHER SCHOOL OF ECONOMICS

DEPARTMENT
CHAIR

PROJECT PROPOSAL

ANALYSIS AND FORECASTING OF MEDICAL EQUIPMENT IMPORTS INTO THE RUSSIAN FEDERATION IN THE CONTEXT OF SANCTIONARY RESTRICTIONS

Ivan Andreev, БЭАД223
Advisor:
Tatyana Ratnikova, PhD
Associate Professor,
Department of Applied Economics

MOSCOW
2026

Despite the absence of direct sanctions on medical equipment, the sector has been significantly affected by secondary constraints due to restrictions on logistics and the departure of Western manufacturers. Since 2022, the Federal Customs Service of Russia has not published detailed foreign trade statistics for the Russian Federation. As a result, there is a lack of statistical studies on the import of medical equipment at the commodity-item and country levels after 2022. Therefore, I collected a mirror trade dataset by parsing data from the statistical agencies and customs services of key partner countries. The first part of the study is focused on identifying the sanctions effect through the Synthetic Control Method, proxy sanctions indices, and an analysis of changes in the structure of imports of medical equipment to Russia using panel data and time-series models. The second part of the study focuses on forecasting the volumes of medical equipment imports using machine-learning and deep-learning models such as Gradient Boosting, Random Forest, and GNNs.

Introduction

Background

According to the Eurasian Economic Commission (EEC, 2022), the largest exporters of medical equipment to Russia in 2021 were China, Germany, the United States, Japan, and the Republic of Korea. Except for China, all of these countries are currently included in the Russian government's list of states that carry out unfriendly actions against the Russian Federation. Despite the absence of direct sanctions on medical equipment exports to Russia, supplies have been significantly constrained due to difficulties in processing payments, the refusal of a number of Western manufacturers to cooperate with Russian counterparties, and logistical disruptions (Pogarskaya, 2025).

Medical devices play a critical role in the healthcare system. Supply disruptions and price growth can undermine the availability of medical services

and lead to the risk of shortages. In this context, analyzing and forecasting medical equipment imports is essential for quantifying these risks.

Problem Statement

This study aims to provide a comprehensive analysis of structural changes in medical equipment imports into Russia under sanctionary restrictions and to forecast import volumes. To achieve this, the following tasks are set:

1. To assess the diversification of import flows across groups of suppliers in the post-sanctions period.
2. To determine the geographical redirection of trade routes through the growth and decline of import volumes by country group.
3. To evaluate how the unit value of supplies has changed across country blocs since 2022.
4. To assess the impact of sanctions on the volatility of trade flows.
5. To identify the heterogeneity of the sanctions effect across medical equipment categories.
6. To identify the main factors affecting medical equipment import volumes in order to develop forecasting models.
7. To compare alternative preprocessing approaches and forecasting models, including machine-learning and deep-learning methods.

Professional Significance

Since 2022, the Federal Customs Service has not published detailed statistics on Russia's foreign trade. Under these conditions, conducting econometric studies of sanctions effects has become difficult due to the lack of up-to-date official data on import volumes by product group and partner country. To solve this problem, a unique monthly mirror-trade dataset on medical equipment imports for 2019-2025 was compiled by parsing data from the customs services of most of Russia's key partner countries in this segment. This study is based on the collected data and applies both classical and modern

methods for econometric analysis of sanctions effects and for forecasting import flows in foreign trade statistics.

Delimitations of the Study

This study is focused on medical equipment imports to Russia and does not address domestic production, government procurement, final consumption, or the availability of medical care.

The dataset contains information on medical equipment imports at the 4-digit level of the Harmonized System (HS), as defined by the World Customs Organization (WCO). A more disaggregated product classification would substantially reduce country coverage, as partner countries publish trade statistics at different levels of aggregation. Informal imports of medical equipment are outside the scope of the analysis.

The study covers the 2019-2025 period. Earlier periods are not included to preserve data comparability and maintain a focus on the current sanctions regime.

The analysis includes Russia's key trading partners for which mirror data could be collected. However, some important exporters of medical equipment to Russia, such as Belarus and Uzbekistan, do not officially publish detailed foreign trade statistics and were therefore excluded from the analysis.

Literature Review

Medical Equipment Imports to Russia

The large-scale sanctions package imposed on Russia in 2022 led to disruptions in medical equipment supplies. As Pogarskaya (2025) notes, unfriendly states have not formally imposed a general ban on exports of medical devices to Russia, but some categories have been included in the dual-use control list. Western firms have refused to cooperate with Russian counterparties due to concerns about secondary sanctions and licensing issues.

Under these conditions, it is important to assess the prospects for import substitution. According to Shtanko and Panikarova (2025), domestic production of medical devices increased by 2.5 times by 2023 compared with the level of 2018. The authors attribute this growth to an expansion of government procurement in the sector, as public clinics and hospitals are the main consumers of Russian medical equipment. However, public procurement is focused on minimizing costs, which leads to a deterioration in product quality. Bizyuk and Andrew (2023) emphasize the urgency of the problem of import substitution in the high-technology medical equipment segment.

The limits of import substitution in high-technology medical equipment have been partially offset by parallel imports authorized by the Government of the Russian Federation in 2022. Tonkonog (2024) explains that importers can supply medical equipment included in the list of the Ministry of Industry and Trade to Russia without the consent of the rights holder. However, Tonkonog (2025) notes that all medical devices entering the market must undergo a process of state registration with confirmation of their safety. The specifics of regulating medical equipment imports into Russia lead to delivery delays and market instability.

Foreign Trade Modelling

The main modern approach to the analysis of bilateral foreign trade is the family of gravity models. Head and Mayer (2013) present the general form of the model as:

$$X_{ni} = GS_i M_n \phi_{ni}$$

In this specification, the model includes the economic size of both the importer (M_n) and the supplier country (S_i), as well as interaction opportunities and market accessibility (ϕ_{ni}) as a multiplicative function. Log-linearization of the model allows the estimation of proportional effects. The authors emphasize that the main empirical value of gravity models lies in the estimation of trade barriers between countries and in accounting for multilateral resistance defined

by Anderson and van Wincoop (2003), which is the impact of the current structure of the global trade network on a given bilateral trade flow.

Practical recommendations on the application of gravity models are described by Yotov et al. (2016). The authors note the importance of properly integrating multilateral resistance into the model and recommend the use of time-varying fixed effects. They also stress the problem of inability to account for zero trade flows of log-linear specification. The authors suggest the Poisson Pseudo-Maximum Likelihood (PPML) estimator as an alternative.

This approach was formalized by Santos Silva and Tenreyro (2006). In addition to the problem of zero values of the dependent variable, the authors show that log-linear estimation leads to biased results in the presence of heteroskedasticity. In this case, the authors propose to estimate the model in multiplicative form using PPML:

$$E[y|X] = e^{X\beta}$$

Sanctions Effect

Quantifying the sanctions effect is a central task of this study. Borisov, Popova, and Rasoulinezhad (2020) examine the net effect of the 2014 political shock on Russia's foreign trade. They include macroeconomic factors for Russia and its trading partners, common external-shock variables, alliance-related and geographical dummy variables in the model specification. The authors assume that Russia's trade flows with EAEU countries are affected only by economic shocks, while the differential effect for other country groups is considered the effect of political shocks. According to the results of the study, the share of trade with sanctioning countries declined, and the effect was heterogeneous across product groups.

Emlinger and Lefebvre (2025) estimate the sanctions effect of the 2022 shock on Russia's trade flows. The authors also use a log-linear gravity specification with a time dummy for February 2022. They show that imports from sanctioning countries declined. There is also an average increase of 3% in

transportation costs with friendly countries, estimated through the ratio of CIF and FOB values.

Yalcin et al. (2025) estimate a gravity model in a multiplicative form to identify the sanctions effect:

$$X_{ij,t} = \exp\left[\mathrm{SANCT}_{\mathrm{ij,t}}\alpha + \mathrm{GRAV}_{\mathrm{ij,t}}\beta + \pi_{\mathrm{i},t} + \chi_{\mathrm{j,t}} + \overrightarrow{\mu_{\mathrm{ij}}}\right] \times \epsilon_{ij,t},$$

It includes common macroeconomic variables for partner countries ($\mathrm{GRAV}_{\mathrm{ij,t}}$), controls for multilateral resistances through time-varying fixed effects of exporters and importers ($\pi_{\mathrm{i},t}$ and $\chi_{\mathrm{j,t}}$), and captures trade barriers through directional country-pair fixed effects ($\overrightarrow{\mu_{\mathrm{ij}}}$). It is estimated using PPML. The authors update Global Sanctions Data Base Release 3 (GSDB-R3), proposed by Syropoulos et al. (2024), through 2023 to account for new sanctions against Russia. The sanctions block ($\mathrm{SANCT}_{\mathrm{ij,t}}$) consists of different types of time-varying sanctions activity (financial, trade, and others). The authors note that trade flows with sanctioning countries decreased, although the effect on these countries is heterogeneous.

Another modern method of identifying sanctions effects in foreign trade is the Synthetic Control Method (SCM), developed by Abadie and Gardeazabal (2003). The effect of a shock is estimated as the difference between the observed trajectory of the treated unit and a weighted combination of the trajectories of donor units. They resemble the behavior of the treated unit before the shock.

Bove, Di Salvatore, and Nisticò (2023) quantify the sanctions effect on the neighbors of a sanctioned country as:

$$\alpha_{0t} = E[Y_{0t}|D_{0t} = 1] - E[Y_{0t}|D_{0t} = 0]$$

$$E[Y_{0t}|D_{0t} = 0] = \sum_{i=1}^{G} w_i \overline{Y_{it}}, \quad \sum_{i=1}^{G} w_i = 1$$

The authors use SCM, arguing that traditional gravity models produce highly aggregated estimates. The results show that the sanctions effect among

the neighbors of a sanctioned country ($D_{0t} = 1$) is indeed highly heterogeneous. On average, the trade flows of neighboring countries decline.

Forecasting

In recent studies, machine-learning methods are used to forecast trade flows. Tiits et al. (2024) note that traditional gravity models are limited for trade forecasting across product groups. These limitations are explained by multicollinearity and instability of estimates. Therefore, the authors extend the gravity approach by applying Random Forest (RF) models. RF predictions are generated by averaging the predictions of multiple decision trees trained on bootstrapped samples of observations and features.

One of the most promising directions in trade forecasting is deep learning. Sellami et al. (2024) examine the application of Graph Neural Networks (GNNs) to foreign trade forecasting. The authors argue that GNNs better capture nonlinear dependencies and network connections between countries than traditional gravity specifications. They show that predictions of graph-based models also outperform RF models in accuracy.

Methods

To address the research objectives, a mirror-trade dataset on monthly medical equipment imports to Russia for 2019-2025 (HS codes 9018–9022, 9025, 9027, 9030, and 9031) was compiled by parsing official data sources of trading partners.

As control variables, this study uses common economic indicators for gravity models that vary across countries and over time (exchange rates, distance-adjusted logistics indices, and other macroeconomic factors). A country–time measure of sanctions activity between Russia and the sanctioning countries was collected using data parsed from X-compliance.

To test the hypotheses, the study uses various specifications of gravity models. The main specification for estimating the effect of sanctions on value

and physical import volumes and for assessing the restructuring of trade routes is:

$$E[Y_{it}|X] = \exp\left[\alpha_i + \delta_t + Sanct_{it}\beta + \sum_g \gamma_g\left(Group_{gi} \cdot Post_t\right) + \mathrm{GRAV}_{it}\theta\right]$$

where Y_{it} denotes import value or import quantity, $Sanct_{it}$ is a proxy for the country's sanctions-related pressure, $Group_{gi} \cdot Post_t$ captures the differential post-shock effect across country alliances, and the GRAV_{it} contains common gravity controls. The model is estimated in multiplicative form using the PPML estimator to avoid problems with zero trade flows and heteroscedasticity. The choice of country and time fixed effects, as well as the use of robust standard errors, is supported by diagnostic tests (F-test, Hausman, and others).

To estimate the sanction effect on non-zero unit values ($UV_{it} = \frac{V_{it}}{Q_{it}}$), a log-linear specification with robust standard errors is used.

The hypothesis of heterogeneity of the sanction effect across product groups is tested using an extended version of the main specification at the HS-code level (α_{ih}) with $\sum_h \lambda_h(Sanct_{it} \cdot HScode_h)$.

The hypothesis of increased volatility in the aggregated time series is tested using Levene and Brown–Forsythe tests as a robust alternative.

The Herfindahl–Hirschman Index (HHI) is used to assess the dependence on a limited number of suppliers:

$$share_{it} = \frac{Q_{it}}{\sum_i Q_{it}}, \qquad HHI_t = \sum_i share_{it}^2$$

Structural changes in concentration are tested using Chow tests and Bai–Perron tests. This approach was recommended by Bacchetta et al. (2012).

As an alternative method for quantifying the sanctions effect, the study uses SCM:

$$\widehat{Y_t} = \sum_{j=1}^{J} w_j Y_{jt}, \qquad w_j \geq 0, \sum_{j=1}^{J} w_j = 1$$

$$\alpha_t = Y_t - \widehat{Y_t},$$

where Y_t is the observed volume of medical equipment imports to Russia, $(\widehat{Y_t})$ is the counterfactual trajectory of donor units.

For the forecasting task, this study compares various methods of preprocessing for continuous variables, categorical features, and seasonality. The best approach to forecasting is identified from linear models (SARIMAX), machine-learning methods (RF and Gradient Boosting), and deep-learning models (GNNs).

Results Anticipated

Based on the results of previous studies, the study expects a decline in import flows from sanctioning countries and an increase in trade with EAEU countries that could be used as transit hubs. The authors also note an increase in the unit cost of imports. The sanctions effect is expected to vary across partner countries and product groups. The largest effect should be anticipated for high-technology medical equipment. The political shock may also affect the stability of supplies. The volatility of flows is expected to increase in the post-sanctions period.

Graph-based models have shown the best performance in forecasting trade flows in previous research. This study also expects an improvement in predictive accuracy when shifting from econometric baselines to machine-learning and deep-learning methods.

Conclusion

In conclusion, this study examines how the 2022 political shock affected the structure of medical equipment imports to Russia. Indirect sanctions-related constraints could have led to higher import costs, delivery delays and an increased risk of shortages. These consequences have a complex impact on the quality of medical care provided to the population. Based on high-frequency data on medical equipment imports, the sanctions effect is quantified with panel-data methods and SCM. The study sequentially tests hypotheses about the growth in import costs and the decline in physical volumes from sanctioning countries, the heterogeneity across countries and product groups, the increase in dependence on a limited number of suppliers, and changes in the volatility of trade flows. Finally, the most effective preprocessing and modelling methods in terms of predictive performance are identified.

References

Abadie, A., & Gardeazabal, J. (2003). *The economic costs of conflict: A case study of the Basque Country*. American Economic Review, 93(1), 113-132.

Anderson, J. E., & van Wincoop, E. (2003). *Gravity with gravitas: A solution to the border puzzle*. American Economic Review, 93(1), 170-192.

Bacchetta, M., Beverelli, C., Cadot, O., Fugazza, M., Grether, J.-M., Helble, M., Nicita, A., & Piermartini, R. (2012). *A practical guide to trade policy analysis*. Geneva, Switzerland: World Trade Organization and United Nations Conference on Trade and Development.

Borisov, A., Popova, L., & Rasoulinezhad, E. (2020). *Modification of Russia's trade pattern under external shocks in 2014 and beyond*. Journal of Economic Integration, 35(2), 296-325.

Bove, V., Di Salvatore, J., & Nisticò, R. (2023). *Economic sanctions and trade flows in the neighbourhood*. Journal of Law and Economics, 66(4), 671-697.

Emlinger, C., & Lefebvre, K. (2025). *Working around sanctions: What cost to Russia?* (No. 2025-50). Paris, France: CEPII research center.

Head, K., & Mayer, T. (2013). *Gravity equations: Workhorse, toolkit, and cookbook*. In G. Gopinath, E. Helpman, & K. Rogoff (Eds.), Handbook of international economics (Vol. 4, pp. 131-195). Amsterdam, Netherlands: Elsevier.

Santos Silva, J. M. C., & Tenreyro, S. (2006). *The log of gravity*. Review of Economics and Statistics, 88(4), 641-658.

Sellami, B., Ounoughi, C., Kalvet, T., Tiits, M., & Rincon-Yanez, D. (2024). *Harnessing graph neural networks to predict international trade flows*. Big Data and Cognitive Computing, 8(6), 65.

Syropoulos, C., Felbermayr, G., Kirilakha, A., Yalcin, E., & Yotov, Y. V. (2024).

The global sanctions data base–Release 3: COVID-19, Russia, and multilateral sanctions. Review of International Economics, 32(1), 12-48.

Tiits, M., Kalvet, T., Ounoughi, C., BenSassi, I., & Yahia, S. B. (2024). *Relatedness and product complexity meet gravity models of international trade*. Journal of Open Innovation: Technology, Market, and Complexity, 10(2), 100288.

Yalcin, E., Felbermayr, G., Kariem, H., Kirilakha, A., Kwon, O., Syropoulos, C., & Yotov, Y. V. (2025). *The Global Sanctions Data Base - release 4: the heterogeneous effects of the sanctions on Russia*. The World Economy, 48(9), 2003-2017.

Yotov, Y. V., Piermartini, R., Monteiro, J.-A., & Larch, M. (2016). *An advanced guide to trade policy analysis: The structural gravity model*. Geneva, Switzerland: World Trade Organization.

Бизюк, Я. И., & Эндрю, Г. Ф. (2023). *Влияние внешнеторговой политики на российский рынок медицинского оборудования*. Российский внешнеэкономический вестник, 12, 101-110.

Евразийская экономическая комиссия. (2022). *Рынок медицинских изделий и изделий, применяемых при оказании медицинской помощи ЕАЭС (2018–2021 гг.)*. Москва, Россия: Евразийская экономическая комиссия.

Погарская, А. С. (2025). *Обзор санкционных мер недружественных стран в отношении экспорта и реэкспорта медицинских изделий в Россию.* Проблемы социальной гигиены, здравоохранения и истории медицины, 33(3), 414-423.

Тонконог, В. В. (2024). *Актуальные вопросы импорта медицинских изделий и оборудования в Российскую Федерацию.* Проблемы социальной гигиены, здравоохранения и истории медицины, 32(S2), 1176-1180.

Тонконог, В. В. (2025). *Особенности таможенного оформления*

медицинских изделий при ввозе в Российскую Федерацию. Проблемы социальной гигиены, здравоохранения и истории медицины, 33(1), 65-72.

Штанько, А. Ю., & Паникарова, С. В. (2025). *Государственное регулирование российского рынка медицинских изделий.* Вопросы государственного и муниципального управления, (4), 139-159.